

Lecture 2 Transform

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Spatial Operation

- ▶ Single pixel operation.
Example: find the complement of image
- ▶ Neighborhood Operations.
Example: Local averaging
- ▶ Geometric Transformation
 - ▶ Spatial transformation of coordinates
 - ▶ Intensity interpolation
- ▶ Image Registration

Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = w$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = c_x v$ $y = c_y w$	
Rotation	$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v \cos \theta - w \sin \theta$ $y = v \sin \theta + w \cos \theta$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$x = v + t_x$ $y = w + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v + s_v w$ $y = w$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = s_h v + w$	

Two Views of Transforms

Transform can be viewed as:

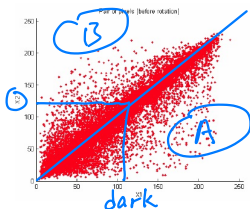
- ▶ Rotation of axes of the original data.
- ▶ Obtaining weights for the basis elements to express input in terms of basis elements

Transform as Rotation of Axes

- Let's look at an example.

Linear Transform

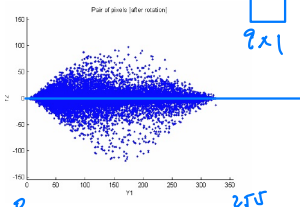
$$Y = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$



before rotation

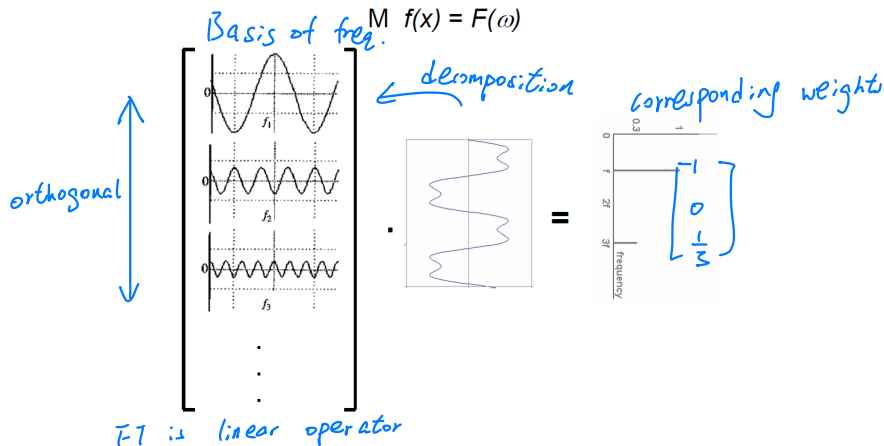


Source Image: Lenna



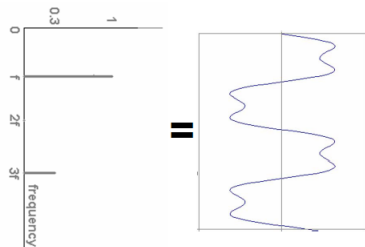
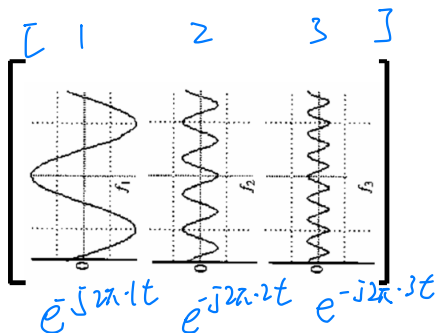
after rotation

FT: Change of basis



IFT: Change of basis

$$M^{-1} F(\omega) = f(x)$$



1D Fourier transform

Reminder transform pair - definition

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx,$$

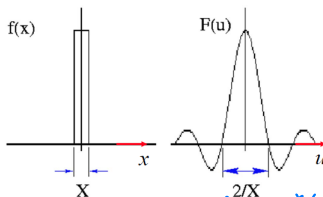
$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du$$

*check specific freq.
⇒ check specific weights*

Example

$$f(x) = \begin{cases} 1, & |x| < \frac{X}{2}, \\ 0, & |x| \geq \frac{X}{2}. \end{cases}$$

$$\begin{aligned} F(u) &= \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx \\ &= \int_{-X/2}^{X/2} e^{-j2\pi ux} dx \\ &= \frac{1}{-j2\pi u} [e^{-j2\pi u X/2} - e^{j2\pi u X/2}] \\ &= X \frac{\sin(\pi Xu)}{(\pi Xu)} = X \text{sinc}(\pi Xu). \end{aligned}$$



$$\int_{-X/2}^{X/2} e^{-j2\pi f x} dx = \frac{e^{-j2\pi f x}}{-j2\pi f} \Big|_{-X/2}^{X/2}$$

2D Fourier transform

Definition

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \boxed{f(x, y)} e^{-j2\pi(ux+vy)} dx dy,$$

image $\rightarrow x, y$ will be 0 or 1
base $\leftarrow f$

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

where u and v are spatial frequencies.

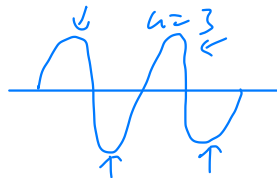
Also will write FT pairs as $f(x, y) \Leftrightarrow F(u, v)$.

- $F(u, v)$ is complex in general,

$$F(u, v) = F_R(u, v) + jF_I(u, v)$$

- $|F(u, v)|$ is the **magnitude** spectrum
- $\arctan(F_I(u, v)/F_R(u, v))$ is the **phase** angle spectrum.
- Conjugacy: $f^*(x, y) \Leftrightarrow F(-u, -v)$
- Symmetry: $f(x, y)$ is **even** if $f(x, y) = f(-x, -y)$

$$F(f) = \int_{-\infty}^{\infty} f(x) e^{j2\pi x f} dx$$



more dense
distribution of
the dark!!!



Sinusoidal Waves

spatial
frequency!
as a
vector

Treat x, y are equally
important!

In 1D the Fourier transform is based on a decomposition into functions $e^{j2\pi ux} = \cos 2\pi ux + j \sin 2\pi ux$ which form an orthogonal basis. Similarly in 2D

$$e^{j2\pi(ux+vy)} = \cos 2\pi(ux+vy) + j \sin 2\pi(ux+vy)$$

The real and imaginary terms are sinusoids on the x, y plane. The maxima and minima of $\cos 2\pi(ux+vy)$ occur when

$$2\pi(ux+vy) = n\pi$$

write $ux+vy$ using vector notation with $\mathbf{u} = (u, v)^T$, $\mathbf{x} = (x, y)^T$ then

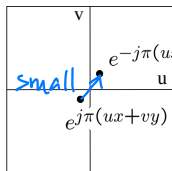
$$2\pi(ux+vy) = 2\pi \mathbf{u} \cdot \mathbf{x} = n\pi$$

no difference!
(inner product!)

are sets of equally spaced parallel lines with normal $\mathbf{u}$ and wavelength $1/\sqrt{u^2+v^2}$.

2D basis for, for this line, periodic!

To get some sense of what basis elements look like, we plot a basis element --- or rather, its real part --- as a function of x, y for some fixed u, v . We get a function that is constant when $(ux+vy)$ is constant. The magnitude of the vector (u, v) gives a frequency, and its direction gives an orientation. The function is a sinusoid with this frequency along the direction, and constant perpendicular to the direction.

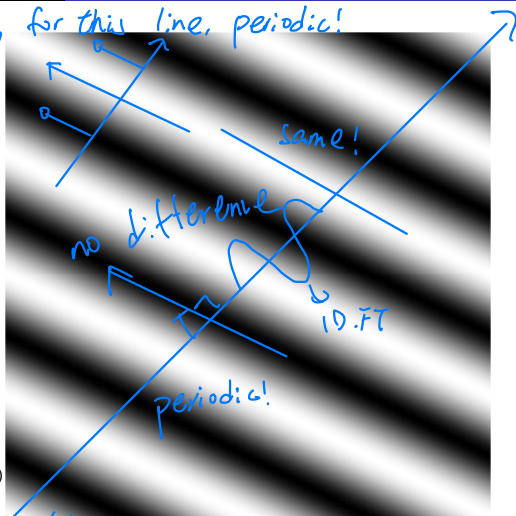


horizontal
&

vertical

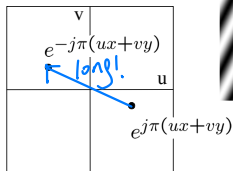
frequency

⇒ want check!



Here u and v are larger than
 in the previous slide.

higher
 frequency!



Transformations

As in the 1D case FTs have the following properties

- **Linearity**

$$\alpha f(x, y) + \beta g(x, y) \Leftrightarrow \alpha F(u, v) + \beta G(u, v).$$

- **Similarity**

$$f(ax, by) \Leftrightarrow \frac{1}{ab} F\left(\frac{u}{a}, \frac{v}{b}\right).$$

This applies, for example, when an image is scaled

- **Shift**

$$f(x - a, y - b) \Leftrightarrow e^{j2\pi(au+bv)} F(u, v)$$

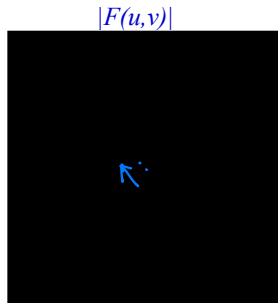
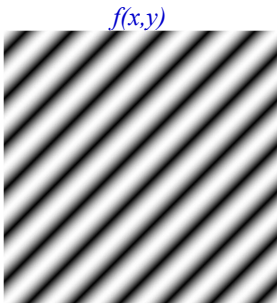
This might apply, for example, if an object moved.

In 2D can also rotate, shear etc

Under an affine transformation: $\mathbf{x} \rightarrow \mathbf{Ax}$ $\mathbf{u} \rightarrow \mathbf{A}^{-\top} \mathbf{u}$

Example

How does $F(u,v)$ transform if $f(x,y)$ is rotated by 45 degrees?



If $\mathbf{A} = \mathbf{R}$ then $\mathbf{A}^{-\top} = \mathbf{R}$.

i.e. FT undergoes the same rotation.

Main Contents

Review on basic mathematical tools

What is Transform?

Fourier Transform

Important Fourier transform Pairs

The convolution theorem

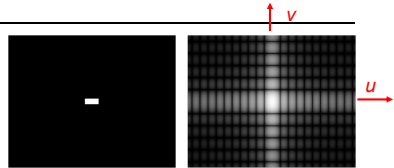
The sampling theorem

Aliasing

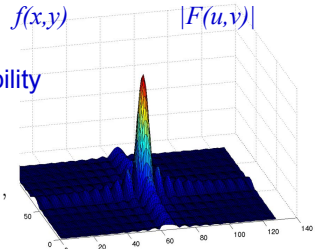
Some important Fourier Transform Pairs

FT pair example 1

rectangle centred at origin
with sides of length X and Y



$$\begin{aligned}
 F(u, v) &= \int \int f(x, y) e^{-j2\pi(ux+vy)} dx dy, \\
 &= \int_{-X/2}^{X/2} e^{-j2\pi ux} dx \int_{-Y/2}^{Y/2} e^{-j2\pi vy} dy, \quad \text{separability} \\
 &= \left[\frac{e^{-j2\pi ux}}{-j2\pi u} \right]_{-X/2}^{X/2} \left[\frac{e^{-j2\pi vy}}{-j2\pi v} \right]_{-Y/2}^{Y/2}, \\
 &= \frac{1}{-j2\pi u} [e^{-juX} - e^{juX}] \frac{1}{-j2\pi v} [e^{-jvY} - e^{jvY}], \\
 &= XY \left[\frac{\sin(\pi Xu)}{\pi Xu} \right] \left[\frac{\sin(\pi Yv)}{\pi Yv} \right] \\
 &= XY \text{sinc}(\pi Xu) \text{sinc}(\pi Yv).
 \end{aligned}$$



$|F(u,v)|$

FT pair example 2

Gaussian centred on origin

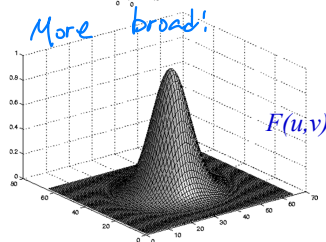
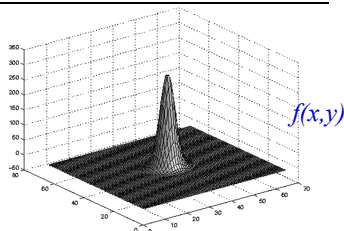
$$f(r) = \frac{1}{2\pi\sigma^2} e^{-r^2/2\sigma^2}$$

where $r^2 = x^2 + y^2$.

$$F(u, v) = F(\rho) = e^{-2\pi^2\rho^2\sigma^2}$$

where $\rho^2 = u^2 + v^2$.

- FT of a Gaussian is a Gaussian
- Note inverse scale relation



FT pair example 3

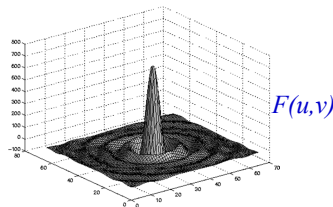
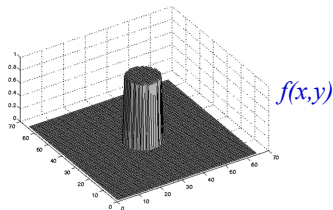
Circular disk unit height and
radius a centred on origin

$$f(x, y) = \begin{cases} 1, & |r| < a, \\ 0, & |r| \geq a. \end{cases}$$

$$F(u, v) = F(\rho) = aJ_1(\pi a \rho) / \rho$$

where $J_1(x)$ is a Bessel function.

- rotational symmetry
- a '2D' version of a sinc



FT pairs example 4

$$f(x, y) = \delta(x, y) = \delta(x)\delta(y)$$

$$\begin{aligned} F(u, v) &= \iint \delta(x, y) e^{-j2\pi(ux+vy)} dx dy \\ &= 1 \end{aligned}$$



$f(x, y)$



$F(u, v)$

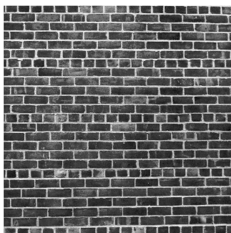
$$f(x, y) = \frac{1}{2}(\delta(x, y - a) + \delta(x, y + a))$$



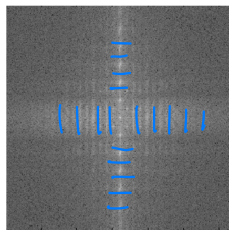
$$\begin{aligned} F(u, v) &= \frac{1}{2} \iint (\delta(x, y - a) + \delta(x, y + a)) e^{-j2\pi(ux+vy)} dx dy \\ &= \frac{1}{2} (e^{-j2\pi av} + e^{j2\pi av}) = \cos 2\pi av \end{aligned}$$

Example 2D Fourier transform

Image with periodic structure



$f(x,y)$



$|F(u,v)|$

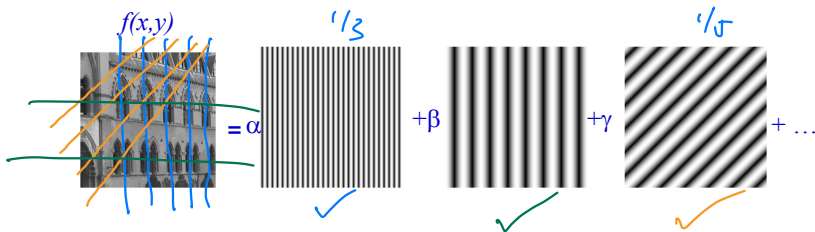
FT has peaks at spatial frequencies of repeated texture

Summary

The spatial function $f(x, y)$

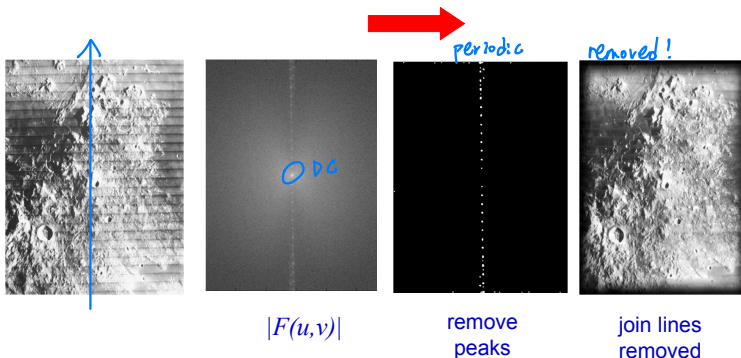
$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

is decomposed into a weighted sum of 2D orthogonal basis functions in a similar manner to decomposing a vector onto a basis using scalar products.



Example – Image processing

Lunar orbital image (1966)



Privacy $\Rightarrow$ blur the faces

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The sampling theorem

Aliasing

The convolution theorem

Filtering vs convolution in 1D

$$g(x) = \sum_i f(x+i)h(i)$$

filtering $f(x)$ with $h(x)$

$f(x)$

100	200	100	200	90	80	80	100	100
-----	-----	-----	-----	----	----	----	-----	-----

$h(x)$

1/4	1/2	1/4
-----	-----	-----

f: filter
molecule/template/kernel

$g(x)$

	150							
--	-----	--	--	--	--	--	--	--

$$g(x) = \int f(u)h(x-u) du$$

convolution of $f(x)$ and $h(x)$

$$= \int f(x+u')h(-u') du'$$

after change of variable $u' = u - x$

$$= \sum_i f(x+i)h(-i)$$

f: flip!

- note negative sign i (which is a reflection in x) in convolution
- $h(x)$ is often symmetric (even/odd), and then (e.g. for even)

$$g(x) = \sum_i f(x+i)h(i)$$

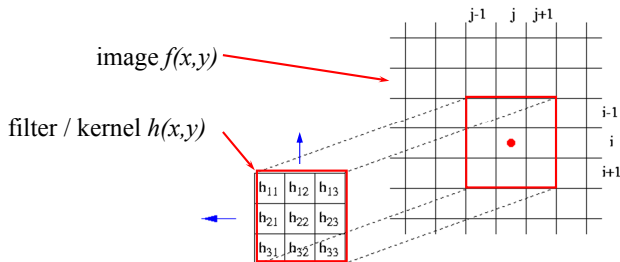
= correlation!

Filtering vs convolution in 2D

convolution

$$\begin{aligned} g(x, y) &= h(x, y) * f(x, y) = f(x, y) * h(x, y) \\ &= \iint f(u, v) h(x - u, y - v) du dv \end{aligned}$$

filtering



$$\begin{aligned} g(x,y) = & h_{11} f(i-1, j-1) + h_{12} f(i-1, j) + h_{13} f(i-1, j+1) + \\ & h_{21} f(i, j-1) + h_{22} f(i, j) + h_{23} f(i, j+1) + \\ & h_{31} f(i+1, j-1) + h_{32} f(i+1, j) + h_{33} f(i+1, j+1) \end{aligned}$$

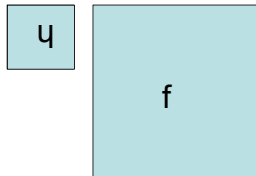
for convolution, reflect filter in x and y axes

Convolution

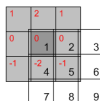
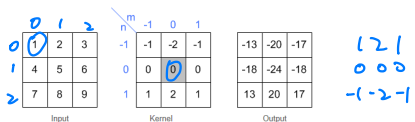
- Convolution: *All position!*
 - Flip the filter in both dimensions (bottom to top, right to left)

$$g[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k h[u, v] f[i - u, j - v]$$

convolution with h



Example of 2D convolution



$$\begin{aligned} y[0,0] &= x[-1,-1] \cdot h[1,1] + x[0,-1] \cdot h[0,1] + x[1,-1] \cdot h[-1,1] \\ &\quad + x[-1,0] \cdot h[1,0] + x[0,0] \cdot h[0,0] + x[1,0] \cdot h[-1,0] \\ &\quad + x[-1,1] \cdot h[1,-1] + x[0,1] \cdot h[0,-1] + x[1,1] \cdot h[-1,-1] \\ &= 0 \cdot 1 + 0 \cdot 2 + 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + 2 \cdot 0 + 0 \cdot (-1) + 4 \cdot (-2) + 5 \cdot (-1) = -13 \end{aligned}$$



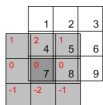
$$\begin{aligned} y[1,0] &= x[0,-1] \cdot h[1,1] + x[1,-1] \cdot h[0,1] + x[2,-1] \cdot h[-1,1] \\ &\quad + x[0,0] \cdot h[1,0] + x[1,0] \cdot h[0,0] + x[2,0] \cdot h[-1,0] \\ &\quad + x[0,1] \cdot h[1,-1] + x[1,1] \cdot h[0,-1] + x[2,1] \cdot h[-1,-1] \\ &= 0 \cdot 1 + 0 \cdot 2 + 0 \cdot 1 + 0 \cdot 0 + 2 \cdot 0 + 3 \cdot 0 + 4 \cdot (-1) + 5 \cdot (-2) + 6 \cdot (-1) = -20 \end{aligned}$$



$$\begin{aligned} y[0,1] &= x[-1,0] \cdot h[1,1] + x[0,0] \cdot h[0,1] + x[1,0] \cdot h[-1,1] \\ &\quad + x[-1,1] \cdot h[1,0] + x[0,1] \cdot h[0,0] + x[1,1] \cdot h[-1,0] \\ &\quad + x[-1,2] \cdot h[1,-1] + x[0,2] \cdot h[0,-1] + x[1,2] \cdot h[-1,-1] \\ &= 0 \cdot 1 + 1 \cdot 2 + 2 \cdot 1 + 0 \cdot 0 + 4 \cdot 0 + 5 \cdot 0 + 0 \cdot (-1) + 7 \cdot (-2) + 8 \cdot (-1) = -18 \end{aligned}$$



$$\begin{aligned} y[1,1] &= x[0,0] \cdot h[1,1] + x[1,0] \cdot h[0,1] + x[2,0] \cdot h[-1,1] \\ &\quad + x[0,1] \cdot h[1,0] + x[1,1] \cdot h[0,0] + x[2,1] \cdot h[-1,0] \\ &\quad + x[0,2] \cdot h[1,-1] + x[1,2] \cdot h[0,-1] + x[2,2] \cdot h[-1,-1] \\ &= 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 1 + 4 \cdot 0 + 5 \cdot 0 + 6 \cdot 0 + 7 \cdot (-1) + 8 \cdot (-2) + 9 \cdot (-1) = -24 \end{aligned}$$



$$\begin{aligned} y[0,2] &= x[-1,1] \cdot h[1,1] + x[0,1] \cdot h[0,1] + x[1,1] \cdot h[-1,1] \\ &\quad + x[-1,2] \cdot h[1,0] + x[0,2] \cdot h[0,0] + x[1,2] \cdot h[-1,0] \\ &\quad + x[-1,3] \cdot h[1,-1] + x[0,3] \cdot h[0,-1] + x[1,3] \cdot h[-1,-1] \\ &= 0 \cdot 1 + 4 \cdot 2 + 5 \cdot 1 + 0 \cdot 0 + 7 \cdot 0 + 8 \cdot 0 + 0 \cdot (-1) + 0 \cdot (-2) + 0 \cdot (-1) = 13 \end{aligned}$$



$$\begin{aligned} y[1,2] &= x[0,1] \cdot h[1,1] + x[1,1] \cdot h[0,1] + x[2,1] \cdot h[-1,1] \\ &\quad + x[0,2] \cdot h[1,0] + x[1,2] \cdot h[0,0] + x[2,2] \cdot h[-1,0] \\ &\quad + x[0,3] \cdot h[1,-1] + x[1,3] \cdot h[0,-1] + x[2,3] \cdot h[-1,-1] \\ &= 4 \cdot 1 + 5 \cdot 2 + 6 \cdot 1 + 7 \cdot 0 + 8 \cdot 0 + 9 \cdot 0 + 0 \cdot (-1) + 0 \cdot (-2) + 0 \cdot (-1) = 20 \end{aligned}$$

Filtering vs convolution in 2D in Matlab

2D filtering

- `g=filter2(h,f);`

`f=image`
`h=filter`

$$g[m,n] = \sum_{k,l} h[k,l] f[m+k,n+l]$$

2D convolution

- `g=conv2(h,f);`

$$g[m,n] = \sum_{k,l} h[k,l] f[m-k,n-l]$$

- The `filter2` function filters data by taking the 2-D convolution of the input `f` and the coefficient matrix `h` rotated 180 degrees

Convolution theorem

$$f(x, y) * h(x, y) \Leftrightarrow F(u, v)H(u, v)$$

Space convolution = frequency multiplication

In words: the Fourier transform of the convolution of two functions is the product of their individual Fourier transforms

Proof: exercise

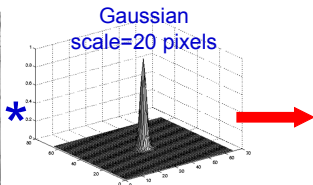
Why is this so important? *(much easier)*

Because linear filtering operations can be carried out by simple multiplications in the Fourier domain

The importance of the convolution theorem

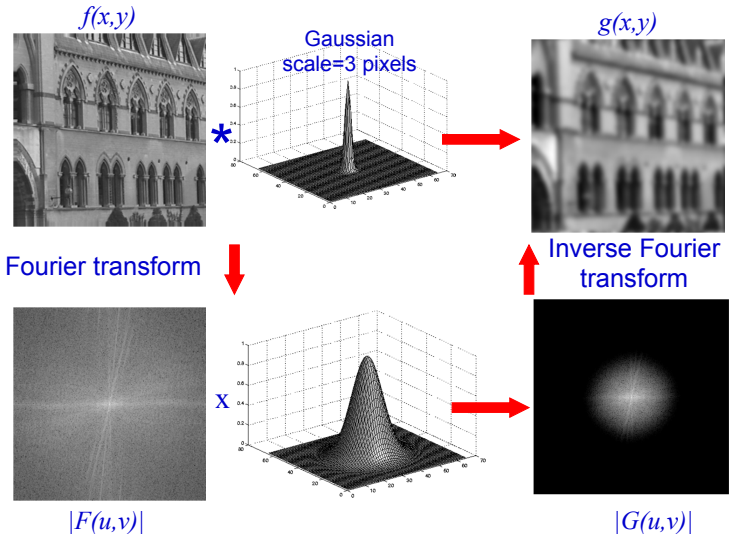
It establishes the link between operations in the frequency domain and the action of linear spatial filters

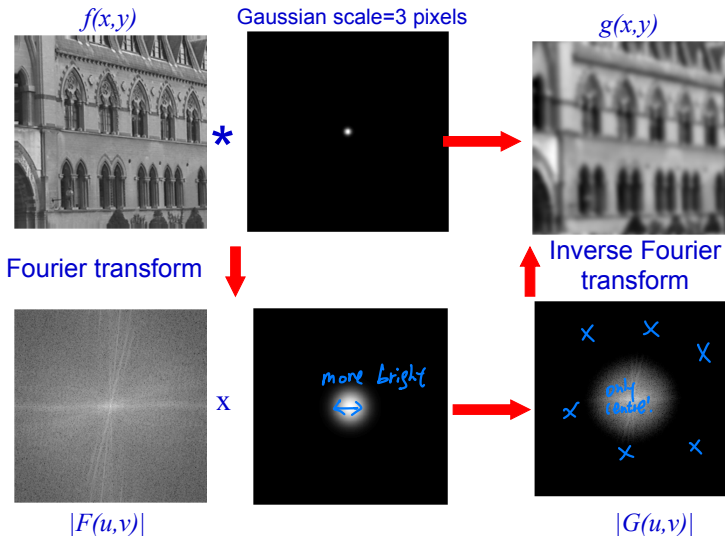
Example smooth an image with a Gaussian spatial filter



1. Compute FT of image and FT of Gaussian
2. Multiply FT's
3. Compute inverse FT of the result.

$$f(x, y) * g(x, y) \Leftrightarrow F(u, v)G(u, v)$$





There are two equivalent ways of carrying out linear spatial filtering operations:

1. Spatial domain: convolution with a spatial operator
2. Frequency domain: multiply FT of signal and filter, and compute inverse FT of product

Why choose one over the other ?

- The filter may be simpler to specify or compute in one of the domains
- Computational cost *↓
easier !*

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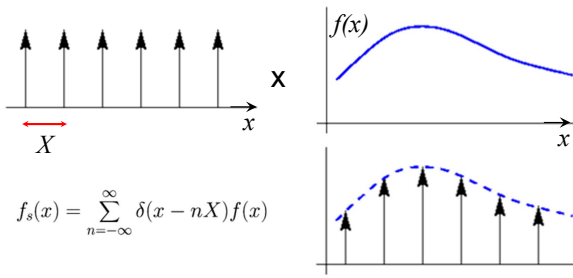
The sampling theorem

Discrete Images - Sampling

In 1D model the image as a set of point samples obtained by multiplying $f(x)$ by the **comb** function

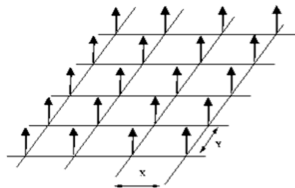
$$\text{comb}(x) = \sum_{n=-\infty}^{\infty} \delta(x - nX)$$

an infinite set of delta functions spaced by X .



In 2D the equivalent of a comb is a **bed-of-nails** function

$$\sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \delta(x - nX) \delta(y - mY)$$

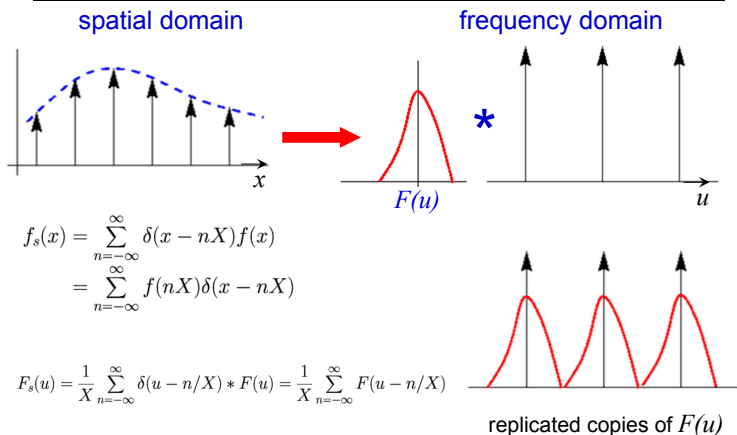


Fourier transform pairs

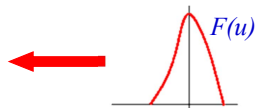
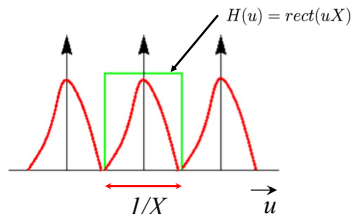
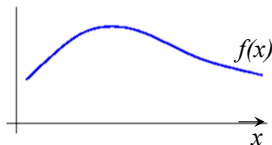
$$\sum_{n=-\infty}^{\infty} \delta(x - nX) \leftrightarrow \frac{1}{X} \sum_{n=-\infty}^{\infty} \delta(u - n/X)$$

$$\sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \delta(x - nX) \delta(y - mY) \leftrightarrow \frac{1}{XY} \sum_{n=-\infty}^{\infty} \delta(u - n/X) \sum_{m=-\infty}^{\infty} \delta(v - m/Y)$$

Sampling Theorem in 1D



Apply a box filter



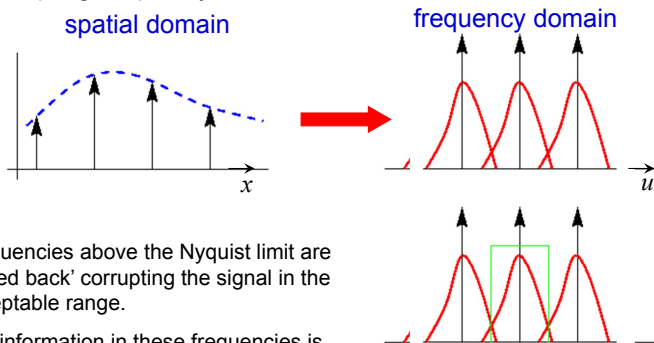
$$F(u) = F_s(u)H(u)$$

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} f(nX)\delta(x - nX) * \text{sinc}\frac{\pi x}{X} \\ &= \sum_{n=-\infty}^{\infty} f(nX)\text{sinc}\frac{\pi}{X}(x - nX) \end{aligned}$$

The original continuous function $f(x)$ is completely recovered from the samples provided the sampling frequency ($1/X$) exceeds twice the greatest frequency of the band-limited signal. (Nyquist sampling limit)

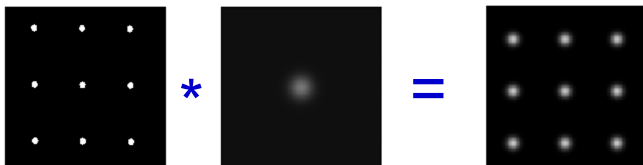
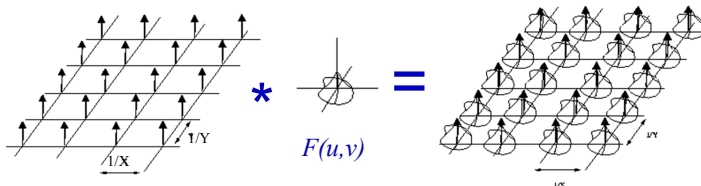
The Sampling Theorem and Aliasing

if sampling frequency is reduced ...



Sampling Theorem in 2D

frequency domain



$$H(u, v) = \text{rect}(uX)\text{rect}(vY)$$

$$f(x, y) = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} f(nX, mY) \text{sinc} \frac{\pi}{X}(x - nX) \text{sinc} \frac{\pi}{Y}(y - nY)$$

The sampling theorem in 2D

If the Fourier transform of a function $f(x,y)$ is zero for all frequencies beyond u_b and v_b , i.e. if the Fourier transform is *band-limited*, then the continuous function $f(x,y)$ can be completely reconstructed from its samples as long as the sampling distances w and h along the x and y directions are such that

$$w < \frac{1}{2u_b} \quad \text{and} \quad h < \frac{1}{2v_b}$$

Main Contents

Review on basic mathematical tools

What is Transform?

Fourier Transform

Important Fourier transform Pairs

The convolution theorem

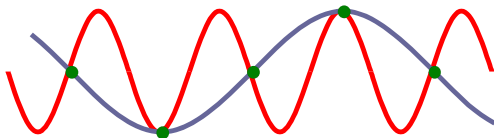
The sampling theorem

Aliasing

Aliasing

Aliasing : 1D example

If the signal has frequencies above the Nyquist limit ...



Insufficient samples to distinguish the high and low frequency

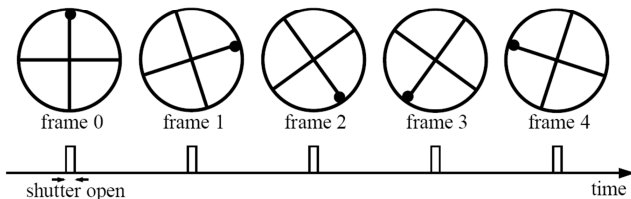
aliasing: signals “travelling in disguise” as other frequencies

Aliasing in video

Imagine a spoked wheel moving to the right (rotating clockwise).

Mark wheel with dot so we can see what's happening.

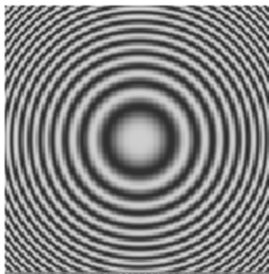
If camera shutter is only open for a fraction of a frame time (frame time = $1/30$ sec. for video, $1/24$ sec. for film):



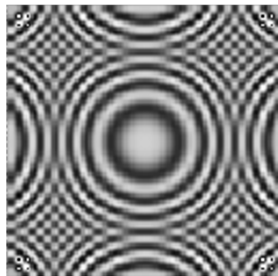
Without dot, wheel appears to be rotating slowly backwards!
(counterclockwise)

Aliasing in 2D – under sampling example

original

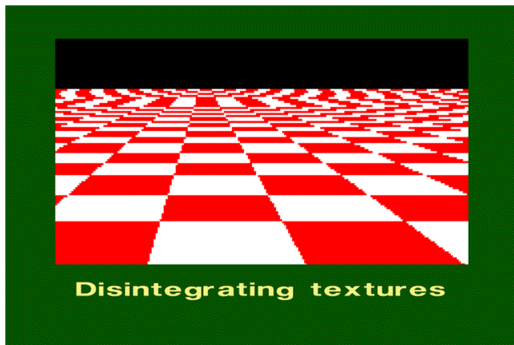


reconstruction



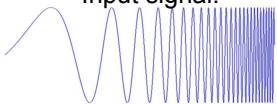
signal has frequencies
above Nyquist limit

Aliasing in images

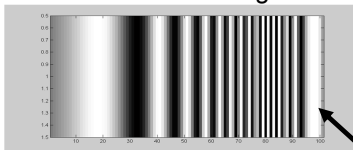


What's happening?

Input signal:



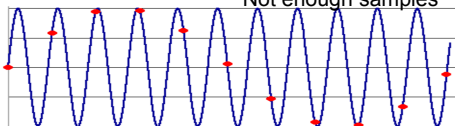
Plot as image:



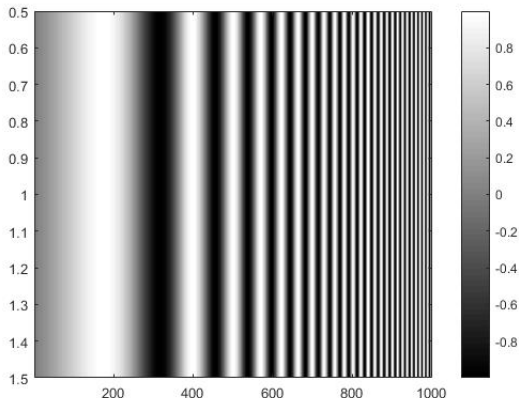
`x = 0:0.05:5; imagesc(sin((2.^x).*x))`

Aliasing

Not enough samples



If you sample enough, $x = [0 : .005 : 5]; \text{imagesc}(\sin((2.^x) .* x));$



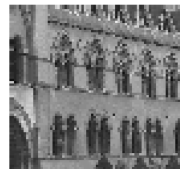
Anti-Aliasing

- Increase sampling frequency
 - e.g. in graphics rendering cast 4 rays per pixel
- Reduce maximum frequency to below Nyquist limit
 - e.g. low pass filter before sampling

Example



down sample by
factor of 4

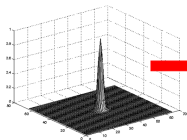


4 x zoom



convolve with
Gaussian

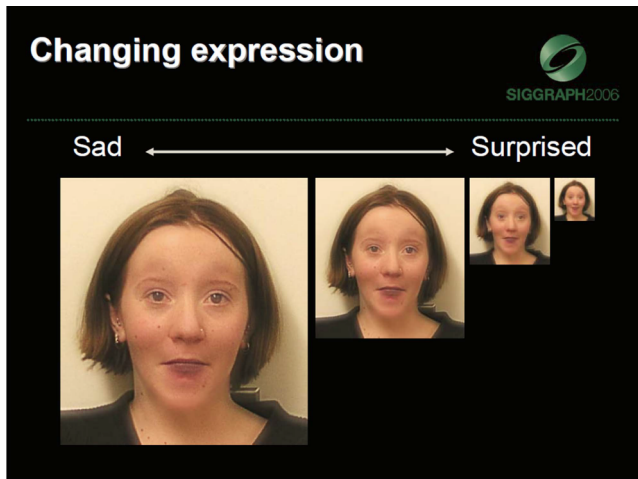
*



down sample by
factor of 4

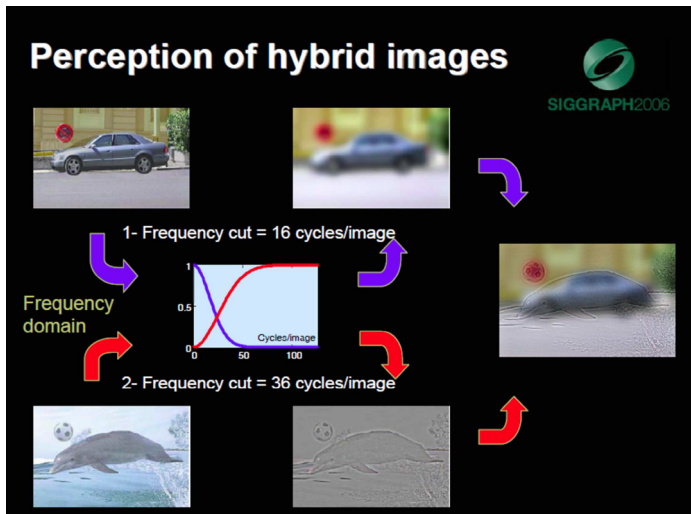


Fun Part: Hybrid Images



Aude Oliva & Antonio Torralba & Philippe G Schyns, SIGGRAPH 2006

How to Generate Hybrid Images



How to play with it?



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Assignment

- ▶ Get yourself familiar with MATLAB.
- ▶ Read about common functions and their Fourier transforms.